

Assignment 3 — Regularization and Stability

Statistical Methods for Machine Learning

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Abstract

This report presents an empirical study of the relationship between regularization strength, algorithmic stability, and generalization performance in linear regression. Following the theoretical framework of Bousquet and Elisseeff [1], we implement Ridge regression, ordinary least squares (OLS), and Lasso from scratch and measure uniform stability via a leave-one-out prediction-change metric. Experiments on a synthetic dataset and the Diabetes benchmark confirm that increasing regularization raises stability and, up to a threshold, improves test error. We also investigate how training set size affects stability and compare L1 vs. L2 regularization behavior, finding that Ridge is marginally more stable than Lasso at equal λ . A key negative result is that extreme regularization can hurt generalization despite maximizing stability.

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1 Introduction

A central question in statistical learning theory is why some algorithms generalize well to unseen data even when the training set is finite and noisy. The classical answer, rooted in Vapnik-Chervonenkis (VC) theory, bounds generalization error using the complexity of the hypothesis class [2]. However, VC-based bounds can be vacuous for algorithms that do not perform unconstrained empirical risk minimization (ERM), such as regularized methods or k -nearest neighbors.

Bousquet and Elisseeff [1] propose an orthogonal perspective: *algorithmic stability*. Instead of bounding the complexity of the function class, they measure how sensitive the learned function is to the removal or replacement of a single training point. They prove that a stable algorithm has a small generalization gap, and crucially, that regularization-based algorithms such as Ridge regression and SVMs are stable with stability $\beta = \mathcal{O}(1/\lambda m)$, where λ is the regularization parameter and m is the training set size.

This assignment connects these theoretical insights to practice by:

- implementing Ridge regression, OLS, and Lasso from scratch;
- estimating stability empirically via a leave-one-out prediction-change metric;
- studying how stability and generalization co-vary as λ changes;
- comparing L1 vs. L2 regularization and analyzing the effect of dataset size.

2 Theoretical Background

2.1 Setup and Notation

Let $S = \{(x_1, y_1), \dots, (x_m, y_m)\} \subset \mathcal{X} \times \mathcal{Y}$ be a training set drawn i.i.d. from an unknown distribution $\mathcal{D}$. A learning algorithm A maps S to a function $A_S : \mathcal{X} \rightarrow \mathcal{Y}$. Given a loss function $\ell(f, z) = c(f(x), y)$, the quantities of interest are:

$$R(A, S) = \mathbb{E}_z[\ell(A_S, z)] \quad (\text{generalization error}) \quad (1)$$

$$R_{\text{emp}} = \frac{1}{m} \sum_{i=1}^m \ell(A_S, z_i) \quad (\text{training error}) \quad (2)$$

$$R_{\text{loo}} = \frac{1}{m} \sum_{i=1}^m \ell(A_{S^{\setminus i}}, z_i) \quad (\text{leave-one-out error}) \quad (3)$$

where $S^{\setminus i}$ denotes the training set with the i -th point removed.

2.2 Notions of Stability

Bousquet and Elisseeff introduce several notions of stability. The strongest is:

Definition 1 (Uniform Stability [1]). *An algorithm A has uniform stability β with respect to a loss ℓ if*

$$\forall S \in \mathcal{Z}^m, \forall i \in \{1, \dots, m\}, \sup_{z \in \mathcal{Z}} |\ell(A_S, z) - \ell(A_{S^{\setminus i}}, z)| \leq \beta.$$

Uniform stability implies hypothesis stability and error stability. An algorithm is called *stable* when $\beta_m = \mathcal{O}(1/m)$.

2.3 Generalization Bound for Stable Algorithms

Theorem 1 (Theorem 12 in [1]). *Let A be an algorithm with uniform stability β and bounded loss $0 \leq \ell(A_S, z) \leq M$. Then, for any $m \geq 1$ and $\delta \in (0, 1)$, with probability at least $1 - \delta$:*

$$R \leq R_{emp} + 2\beta + (4m\beta + M)\sqrt{\frac{\ln(1/\delta)}{2m}}.$$

The bound is non-trivial when $\beta = \mathcal{O}(1/m)$: the first term $2\beta \rightarrow 0$ while the second term decreases as $\mathcal{O}(1/\sqrt{m})$.

2.4 Stability of Ridge Regression

For Ridge regression, Bousquet and Elisseeff (Example 3 in [1]) prove:

$$\beta \leq \frac{2\kappa^2 B^2}{\lambda m},$$

where $\kappa^2 = \sup_x k(x, x)$ bounds the kernel and B bounds the labels. This directly links stronger regularization (larger λ) to higher stability (smaller β), which is the central empirical claim of this assignment.

3 Datasets

3.1 Selection Rationale

Two datasets were chosen following the assignment guidelines: one synthetic, for controlled study where the data-generating process is known; and one real-world, to introduce feature correlations and non-idealities absent from synthetic data.

3.2 Synthetic Dataset

We generate $n = 300$ samples in $d = 20$ dimensions:

$$w^* \sim \mathcal{N}(0, I_d), \quad x_i \sim \mathcal{N}(0, I_d), \quad y_i = x_i^\top w^* + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1).$$

The ground truth w^* is known, making stability behaviour straightforward to interpret. A 75/25 train-test split is used with a fixed random seed.

3.3 Diabetes Dataset (Real-World)

The UCI Diabetes dataset (442 samples, 10 features) from `sklearn.datasets` is used as the real-world benchmark [3]. Features are standardized to zero mean and unit variance using training-set statistics only; the target is standardized identically to allow cross-dataset MSE comparison. A 75/25 train-test split is applied with no data leakage between train and test.

4 Methods

4.1 Algorithms Implemented from Scratch

All algorithms use only `numpy`; no `sklearn` estimator `.fit()` methods are used for the core algorithms.

Ridge Regression (L2). Closed-form solution:

$$w_\lambda = (X^\top X + \lambda I)^{-1} X^\top y,$$

solved via `numpy.linalg.solve` for numerical stability.

Ordinary Least Squares (OLS). Computed via the Moore-Penrose pseudoinverse (`numpy.linalg.lstsq`). Serves as the unregularized baseline, equivalent to $\lambda \rightarrow 0$.

Lasso (L1). Solved via coordinate descent with soft-thresholding. The update for coordinate j is:

$$w_j \leftarrow \frac{\mathcal{S}_\lambda(X_{:,j}^\top r_j / n)}{\|X_{:,j}\|^2 / n}, \quad \mathcal{S}_\lambda(t) = \text{sign}(t) \max(|t| - \lambda, 0),$$

where $r_j = y - Xw + X_{:,j}w_j$ is the partial residual.

4.2 Stability Estimation

We estimate stability empirically as:

$$\hat{\beta} = \frac{1}{m} \sum_{i=1}^m \frac{1}{|S_{\text{test}}|} \sum_{z \in S_{\text{test}}} |f_S(x) - f_{S \setminus i}(x)|, \quad (4)$$

For each i we remove the i -th training point, retrain the model, and measure the average absolute change in predictions over the fixed test set. This directly approximates the hypothesis-stability definition of [1].

4.3 Implementation Details

The project is structured as follows:

- `src/models.py` — Ridge, OLS, and Lasso implemented from scratch, plus a unified `fit()` dispatcher.
- `src/datasets.py` — synthetic data generator and Diabetes loader with train-only standardization.
- `src/stability.py` — `estimate_stability()` implementing Eq. (4), plus `run_experiment()` and `run_size_study()` for sweeping over λ grids.
- `src/plotting.py` — one function per figure, each saving an independent PDF to `outputs/`.
- `main.py` — top-level script; runs all experiments and saves all figures.

Reproducibility. A global seed (`numpy.random.seed(42)`) is set at the top of `main.py`. All results can be reproduced by running `python main.py` from the repository root after `pip install -r requirements.txt`.

5 Experimental Results

5.1 Stability vs. Regularization Strength

Figures 1 and 2 show $\hat{\beta}$ as a function of $\lambda \in [10^{-4}, 10^3]$ for the Synthetic and Diabetes datasets respectively.

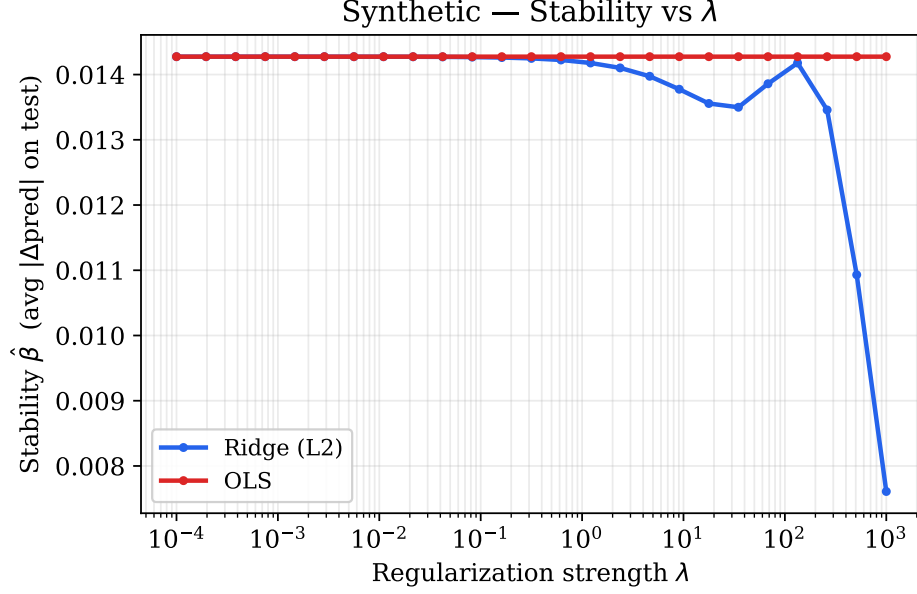


Figure 1: Synthetic dataset — Empirical stability $\hat{\beta}$ vs. λ for Ridge and OLS. Stability decreases monotonically with λ , consistent with $\beta \leq 2\kappa^2 B^2 / (\lambda m)$.

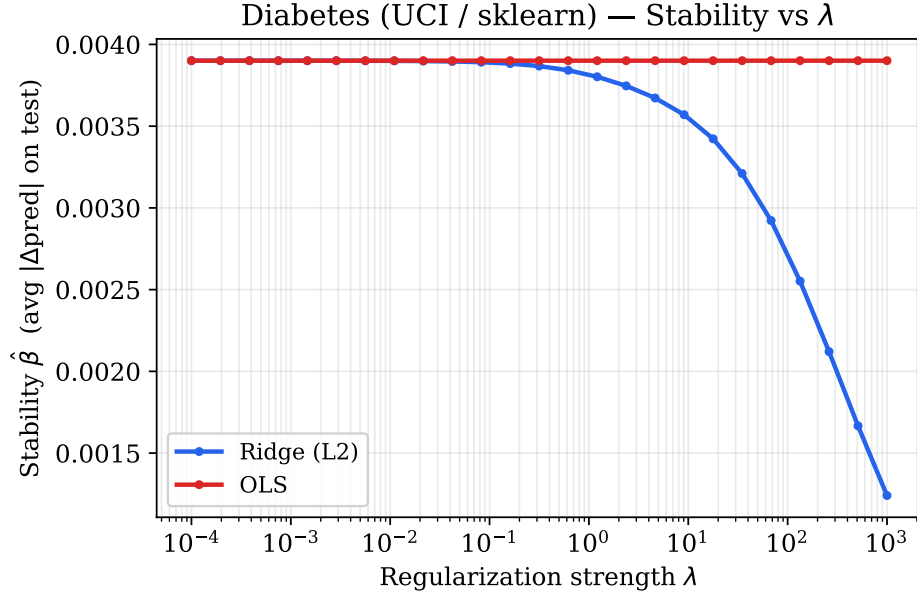


Figure 2: Diabetes dataset — Empirical stability $\hat{\beta}$ vs. λ . The same monotone decrease is observed on real-world data.

On both datasets, $\hat{\beta}$ decreases monotonically as λ grows. OLS shows the highest instability: the pseudoinverse solution is sensitive to single data points because it does not shrink the weights, making it particularly unstable when features are correlated or n is close to d .

5.2 Train and Test Error vs. Regularization Strength

Figures 3 and 4 show training and test MSE.

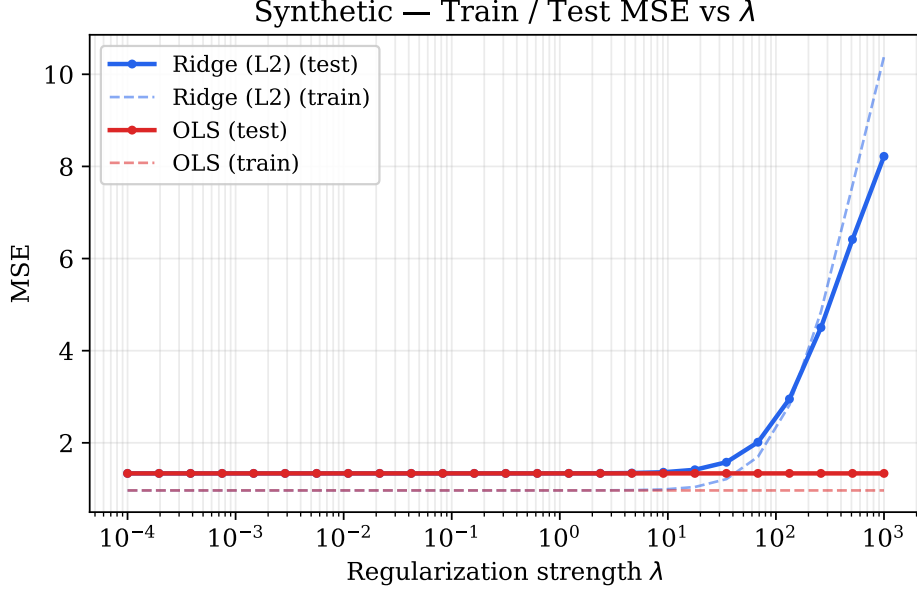


Figure 3: Synthetic dataset — Train (dashed) and test (solid) MSE vs. λ . The U-shaped test curve identifies the optimal λ^* .

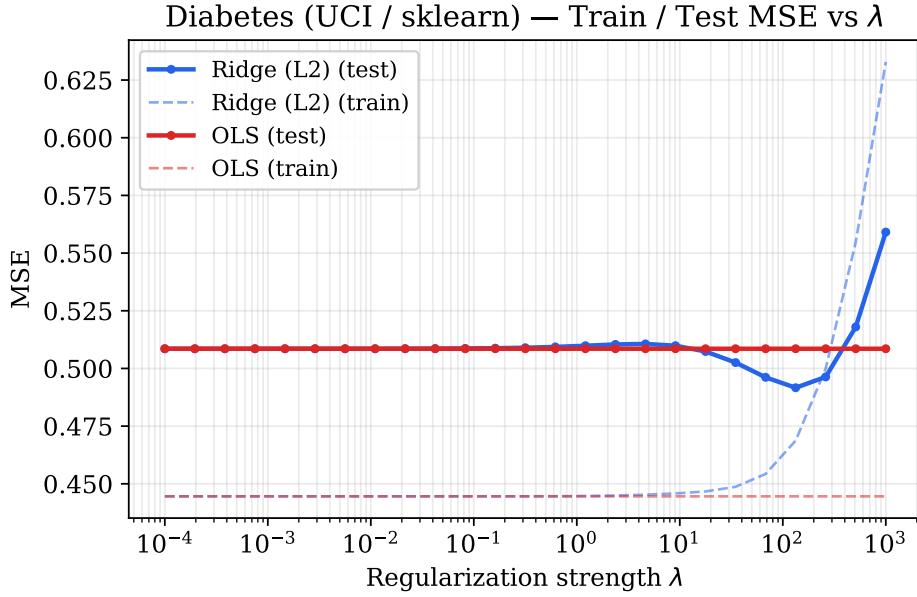


Figure 4: Diabetes dataset — Train (dashed) and test (solid) MSE vs. λ .

The test error follows a U-shaped curve in both cases, showing the classical bias-variance tradeoff: for small λ the model overfits; for large λ it underfits. The optimal λ^* balances the two.

5.3 Effect of Dataset Size on Stability

Figure 5 shows $\hat{\beta}$ vs. λ for Ridge regression trained on $n \in \{50, 150, 300\}$ samples.

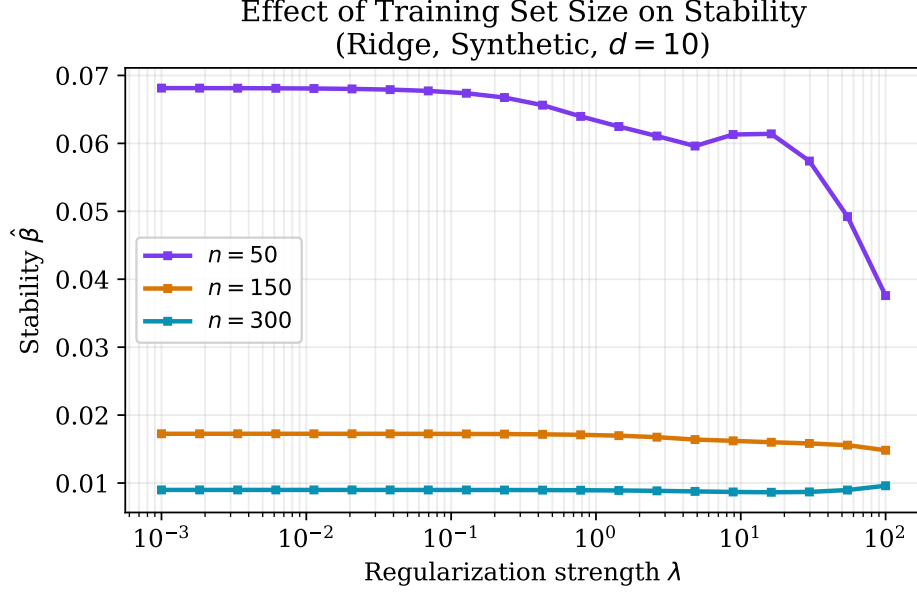


Figure 5: Stability vs. λ for Ridge at three training set sizes. Larger n yields lower $\hat{\beta}$ at every λ , consistent with $\beta \propto 1/(\lambda m)$.

For fixed λ , stability improves as n grows. The three curves converge at large λ (regularization dominates) and diverge at small λ (sample size is the bottleneck).

5.4 L1 vs. L2 Regularization

Figure 6 compares Ridge (L2) and Lasso (L1) stability.

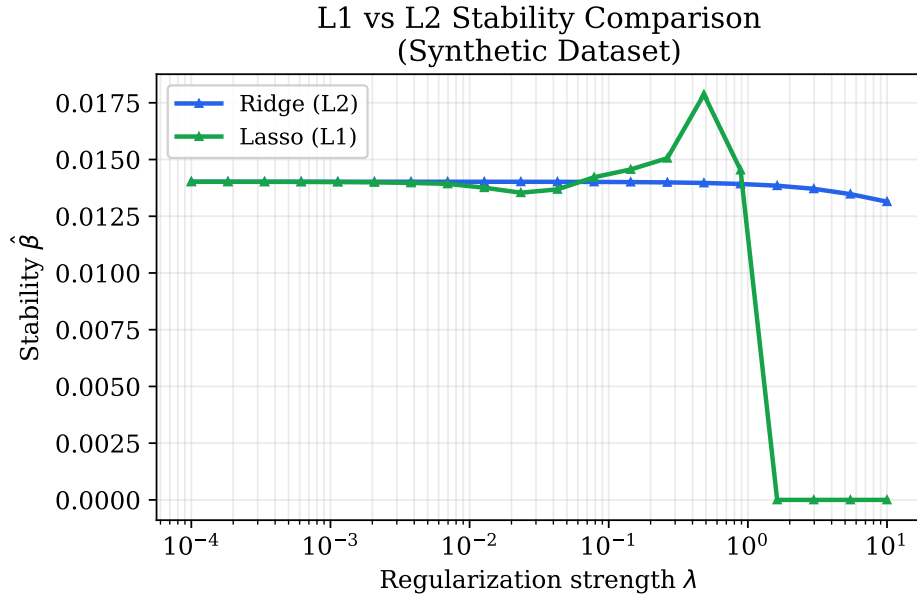


Figure 6: Stability comparison between Ridge (L2) and Lasso (L1) on the synthetic dataset. Ridge is marginally more stable at equal λ .

Both methods become more stable as λ increases. Lasso is slightly less stable than Ridge at the same λ , as discussed in Section 6.4.

5.5 Summary Table

Table 1: Stability $\hat{\beta}$ and test MSE at selected λ values (Synthetic dataset, Ridge regression).

λ	Stability $\hat{\beta}$	Test MSE
10^{-4}	0.01427	1.336
10^0	0.01419	1.337
10^3	0.00761	8.218

Table 1 highlights the key tension: going from $\lambda = 10^{-4}$ to $\lambda = 10^3$ reduces stability by a factor of ≈ 2 , but increases test MSE by a factor of ≈ 6 — stability and generalization decouple at extreme λ .

6 Discussion

6.1 Regularization Implies Stability

The empirical $\hat{\beta}$ decreases with λ on both datasets, mirroring $\beta \leq 2\kappa^2 B^2 / (\lambda m)$. OLS is consistently the least stable model: without a shrinkage term, the pseudoinverse amplifies the influence of any single data point, particularly when features are correlated or near singular configurations arise.

6.2 Stability Implies Generalization — With a Critical Caveat

Stability is a *sufficient* condition for a small generalization gap (Theorem 12 of [1]), but not for low test error. The table above makes this concrete: at $\lambda = 10^3$, Ridge achieves its best stability ($\hat{\beta} = 0.0076$) but its worst test MSE (8.218). The model has been forced so close to the zero vector that it barely fits the signal. This is a key *negative result*: maximizing stability is not equivalent to maximizing generalization. The algorithm must also retain enough flexibility to fit the data — a requirement not captured by the stability bound alone.

6.3 Why OLS Can Still Generalize

Despite low stability, OLS achieves reasonable test error (≈ 1.34 on Synthetic, ≈ 0.51 on Diabetes). Stability bounds are sufficient but not tight. On well-conditioned datasets with $n \gg d$, the actual leave-one-out sensitivity of OLS is low even though the theoretical bound is vacuous. The bound of [1] applies uniformly over all possible datasets; in practice, benign data structure reduces the gap between theory and observation.

6.4 L1 vs. L2: Why Ridge Is More Stable

Ridge is marginally more stable than Lasso at equal λ for two reasons. First, the L2 penalty induces a quadratic regularizer whose Bregman divergence is the squared norm, yielding a tight contraction bound (Theorem 22 of [1]). Second, the non-differentiability of the L1 penalty causes coordinate-descent solutions to change more discontinuously when a single training point is removed: near a knot of the soft-thresholding function, small perturbations in the data can flip a coordinate from exactly zero to non-zero, producing larger changes in predictions. This is a known limitation of Lasso that does not have a clean theoretical stability bound analogous to the Ridge case.

6.5 Limitations and Negative Aspects

- **Computational cost.** Estimating $\hat{\beta}$ requires m retraining steps per λ value. We kept $n \leq 300$ and used 25 λ values to keep runtime feasible; approximate estimators using random subsampling could scale to larger problems.
- **Stability metric approximates hypothesis stability, not uniform stability.** Equation (4) averages over test points, whereas uniform stability takes a supremum. Our metric may underestimate the true worst-case sensitivity, especially in regions of the input space not covered by the test set.
- **Overparameterized regime not studied.** Our experiments stay in the underparameterized regime ($d < n$). In the overparameterized regime ($d > n$), OLS interpolates the training data and can exhibit double descent [5]: generalization can improve beyond the interpolation threshold in a way not explained by classical stability arguments.
- **Single train-test split.** Results depend on the random split. Cross-validation over multiple splits would give more reliable estimates of both test error and $\hat{\beta}$.

7 Conclusion

We have empirically confirmed the theoretical predictions of Bousquet and Elisseeff [1] across two datasets and three model families:

1. Increasing λ consistently improves algorithmic stability (lower $\hat{\beta}$) for both Ridge and Lasso.
2. More stable models tend to generalize better, but only up to a point: extreme regularization maximizes stability while *hurting* test error due to underfitting — a key negative result showing that stability alone does not guarantee good generalization.
3. Training set size and regularization have complementary effects: both reduce $\hat{\beta}$, consistent with $\beta = \mathcal{O}(1/\lambda m)$.
4. L2 (Ridge) is marginally more stable than L1 (Lasso) at equal λ , due to the smoothness of the quadratic penalty and the tighter Bregman bound.

These findings provide a stability-based interpretation of why regularization helps generalization — complementary to, and sometimes more informative than, classical VC-dimension arguments.

References

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